

Canonical commutation relation

Let X and P be the operators

$$X = x \quad P = -i\hbar \frac{\partial}{\partial x}$$

Show that

$$(XP - PX)\psi(x, t) = i\hbar\psi(x, t)$$

Eigenmath code

```
X(f) = x f
P(f) = -i hbar d(f,x)
X(P(psi(x,t))) - P(X(psi(x,t)))
```

Result

$$i\hbar\psi(x, t)$$

In three dimensions (symbol $\otimes$ is outer product, ∇ is gradient)

$$X = \begin{pmatrix} x \\ y \\ z \end{pmatrix} \otimes \quad P = -i\hbar \nabla$$

Eigenmath code

```
X(f) = outer((x,y,z),f)
P(f) = -i hbar d(f,(x,y,z))
X(P(psi(x,y,z,t))) - P(X(psi(x,y,z,t)))
```

Result

$$\begin{pmatrix} i\hbar\psi(x,y,z,t) & 0 & 0 \\ 0 & i\hbar\psi(x,y,z,t) & 0 \\ 0 & 0 & i\hbar\psi(x,y,z,t) \end{pmatrix}$$

In momentum space

$$X = i\hbar \frac{\partial}{\partial p} \quad P = p$$

Eigenmath code

```
X(f) = i hbar d(f,p)
P(f) = p f
X(P(psi(p,t))) - P(X(psi(p,t)))
```

Result

$$i\hbar\psi(p,t)$$

In three dimensions

$$X = i\hbar\nabla_p \quad P = \begin{pmatrix} p_x \\ p_y \\ p_z \end{pmatrix} \otimes$$

Eigenmath code

```
X(f) = i hbar d(f,(px,py,pz))
P(f) = outer((px,py,pz),f)
X(P(psi(px,py,pz,t))) - P(X(psi(px,py,pz,t)))
```

Result

$$\begin{pmatrix} i\hbar\psi(p_x, p_y, p_z, t) & 0 & 0 \\ 0 & i\hbar\psi(p_x, p_y, p_z, t) & 0 \\ 0 & 0 & i\hbar\psi(p_x, p_y, p_z, t) \end{pmatrix}$$

Eigenmath script